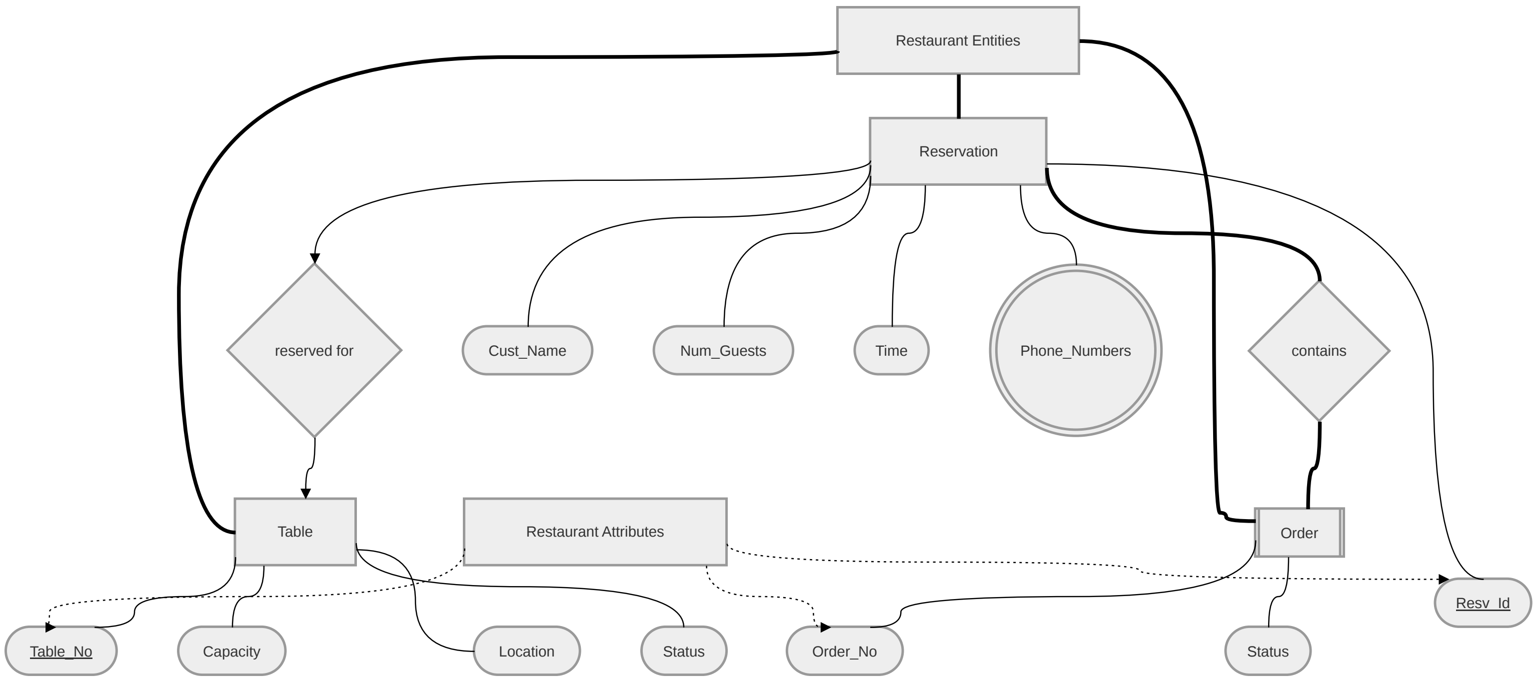
Restaurant Entities
Reservation
reserved for
Cust_Name
Num_Guests
Time
Phone_Numbers
contains
Table
Restaurant Attributes
Order
Table_No
Capacity
Location
Status
Order_No
Status
Resv_Id

ER Diagram Documentation: Design Choices & Assumptions

1. Architectural Focus & Scope

This ER diagram is constructed as a conceptual model focusing on the core operational flow of the restaurant: managing table inventory, seating guests, and processing their requests. To maintain visual clarity and adhere strictly to Chen's Database Notation, secondary logistical elements (such as Waiters and final Bills) were omitted in favor of detailing the precise mechanics of the Reservation, Table, and Order lifecycle.

2. Entity & Relationship Considerations

- **Reservation to Table (reserved for):** This is the central relationship of the system. We model this conceptually to show that a Reservation is assigned to a Table. Over time, a table will have many reservations, but at any specific given time, a reservation is tied to a specific table assignment.
- **Order as a Weak Entity:** We explicitly modeled Order as a Weak Entity (represented by the double rectangle).
  - *Design Choice:* An order cannot logically exist without a reservation or table session to attach it to. If a reservation is canceled or a party leaves, their pending orders are inherently tied to that event.
  - *Identifying Relationship:* We used the "Total Participation" relationship contains (double lines/thick diamond) to show that every order must be linked to a parent reservation.

3. Attribute Considerations

- **Primary Keys (PK):** Resv_Id and Table_No are explicitly marked as primary keys (solid underline) to ensure every booking and physical table can be uniquely identified in the database.
- **Partial Keys:** Because Order is a weak entity, Order_No is marked as a partial key (italicized, dotted underline). By itself, "Order #1" is not unique across the restaurant. It requires the parent key (Resv_Id + Order_No) to form a complete, unique composite key in the database.
- **Multi-Valued Attributes:** We designed Phone_Numbers as a multi-valued attribute (double circle) attached to Reservation.
  - *Design Choice:* Customers frequently provide more than one contact method (e.g., a primary cell phone and a secondary home/work number). Modeling this as multi-valued

ensures the database schema can normalize this properly later (likely creating a separate table for phone numbers) rather than artificially limiting the system to a single Phone column.

4. Business Logic Assumptions

- **Walk-ins vs. Advance Bookings:** We assume the system treats "Walk-in" customers exactly the same as "Advance" reservations. For a walk-in, the host simply generates a new Reservation record at the current Time to track the guest, assign the Table, and link the subsequent Orders.
- **Capacity Management:** We assume Num_Guests (on Reservation) and Capacity (on Table) are tracked separately to allow the system (or host) to validate that a given table can physically accommodate the arriving party before completing the reserved for relationship.